

DATA CLEANING REPORT

Fossil Fuel Support Analysis

Türkiye • 2020–2023

Project	Türkiye Fossil Fuel Support Analysis (2020–2023)
Dataset	Calculated_and_Cleaned_Turkey_FFS_Data.xlsx
Prepared by	Data Analyst
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01 The Problem

Government records of fossil fuel support in Türkiye between 2020 and 2023 were spread across a single dataset with 224 rows of raw information. Before any meaningful analysis could begin, the data needed to be carefully reviewed and verified to ensure the numbers were trustworthy.

Three core problems made the raw data unreliable for direct analysis:

- Hidden sub-records: Many rows that appeared to be identical were actually separate legitimate entries for the same support program in the same year. Treating these as errors and removing them would have cut the true total support value roughly in half.
- Unreliable pre-calculated columns: Several figures in the dataset , including year-on-year percentage changes, monetary impact labels, trend labels, and average annual support , were computed before the analyst received the data. The method behind those calculations was unknown, meaning they could not be trusted without independent verification.
- A currency trap: All financial figures were recorded in Turkish Lira (TRY). Turkey's currency lost significant value between 2020 and 2023, with annual inflation running as high as 72% in 2022. Comparing raw lira figures across years without accounting for this gives a misleading picture of whether support was actually growing or shrinking.

Without addressing these three issues, any chart or summary produced from the data would likely tell the wrong story.

02 Aim

The aim of this data cleaning exercise was to produce a verified, analysis-ready version of the dataset that could be used with confidence to answer the project's key business questions.

Specifically, the cleaning process set out to:

- Understand what is actually in the data before making any changes , row count, column types, missing values, and category breakdowns.
- Identify which rows were true duplicates versus legitimate sub-records that needed to be kept and summed.
- Verify or recalculate any pre-computed columns rather than accepting them at face value.
- Create three parallel versions of each financial figure: one in nominal Turkish Lira as originally reported, one adjusted for inflation into equivalent 2023 Turkish Lira, and one converted to US Dollars using the exchange rate for each year.
- Flag any unusual or extreme values that warranted attention before analysis.

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Project Overview

This project was commissioned by a multinational energy analytics firm to evaluate how the Turkish government distributed financial support to fossil fuel industries between 2020 and 2023. The goal is to produce insights that can inform policy decisions, investment strategies, and sustainability reporting.

Turkey is an unusual case in energy policy: the country imports approximately 93% of its oil and 99% of its natural gas. This means that government support is not primarily aimed at boosting domestic production , it is aimed at shielding households and businesses from the price shocks that come with relying so heavily on international energy markets.

The four years covered by the dataset were also unusually turbulent. The COVID-19 pandemic suppressed global energy demand in 2020. A sharp recovery in 2021 pushed prices upward. Russia’s invasion of Ukraine in February 2022 sent European gas prices to all-time highs. And throughout this period, the Turkish lira was losing value rapidly due to a combination of high inflation and unconventional economic policy.

Why this context matters for data cleaning

Understanding the external environment explains why the raw numbers look the way they do. A jump in support spending in 2022 is not a data error , it is a reflection of the Ukraine war shock. A figure that appears stable in nominal lira may represent a dramatic real decline once inflation is accounted for. Context is what separates a cleaning decision from a cleaning mistake.

Dataset at a Glance

Attribute	Detail
Total rows	224
Columns	17
Years covered	2020, 2021, 2022, 2023
Rows per year	56 (perfectly balanced)

Attribute	Detail
Unique support programs	26
Fossil fuel categories	5 (Oil, Fuel, Coal, Natural Gas, General)
Support types	2 (Tax Relief, Other Support Type)
Missing values	26 (YoY_Change column only , expected base-year rows)
Currency	Turkish Lira (TRY)

04 Problems Identified

Problem 1: Apparent Duplicates That Were Not Duplicates

The most consequential issue in the dataset was the presence of rows that shared the same year and instrument code. At first glance, these looked like data entry errors , the same program appearing twice or more in the same year. In fact, 188 out of 224 rows were part of these overlapping groups.

Closer inspection of the actual program descriptions revealed that these were legitimate sub-records: individual line items within a single support program that needed to be summed to get the true total. The program TUR_DT_01 (Aid to the Hard Coal Industry) in 2020, for example, split cleanly into two entries of ₺921 million and ₺579 million, totalling ₺1.5 billion. Deleting either entry would have silently understated the program's value.

Year+Code group size	Number of groups	Action taken
1 row (unique)	36 groups	Kept as-is
2 rows	44 groups	Summed to single record
3 rows	12 groups	Summed to single record
4 rows	4 groups	Summed to single record
6 rows	8 groups	Summed to single record

After correctly aggregating sub-records, the 224-row dataset consolidated to 104 summarised records , one per program per year , which is the correct analytical unit.

Problem 2: Unreliable Pre-Calculated Columns

The dataset arrived with several columns that had already been computed by a prior process. Because the assumptions behind those calculations were not documented, they could not be trusted without independent verification. The affected columns were:

- **YoY_Change:** Year-on-year percentage change. This column had 26 missing values in base-year rows, which is expected, but several extreme values suggested calculation errors , particularly in rows where the value 703,716,490 appeared to have been used as a placeholder (see Problem 4 below).

- **Monetary_Impact:** A label (Low, Medium, High) for each row. The thresholds used to assign these labels were not documented, making the labels potentially arbitrary.
- **Average_Annual_Support:** A single average figure applied identically to all years of each program. Since this was calculated across the full four-year period, it masked year-to-year variation.
- **Trend:** Labels of Increasing, Decreasing, or Stable. These were derived from the nominal figures without inflation adjustment, making them potentially misleading.

Decision

All four columns were recalculated independently by the analyst. Year-on-year changes were recalculated from the verified summed values. Monetary impact thresholds were documented and re-applied consistently. Trend labels were recalculated on both nominal and inflation-adjusted values.

Problem 3: Nominal vs Real Currency , The Inflation Trap

All financial figures in the dataset were recorded in nominal Turkish Lira: the face value of money at the time it was spent. The problem is that Turkey experienced severe inflation across the four years covered. In 2021, prices rose 19.6%. In 2022, they rose 72.3%. In 2023, they rose 53.9%.

When prices rise this fast, a lira spent in 2020 buys far more than a lira spent in 2023. If you compare 2020 and 2023 figures in nominal terms without adjusting for inflation, you will almost certainly conclude that support was growing. The honest comparison tells a completely different story.

Year	Annual Inflation	Deflator Applied	Effect
2020	~12.3%	× 2.70	2020 figures multiplied up to 2023 equivalent
2021	19.6%	× 2.25	2021 figures multiplied up to 2023 equivalent
2022	72.3%	× 1.47	2022 figures multiplied up to 2023 equivalent
2023	53.9%	× 1.00	Base year , no adjustment needed

Once these deflators are applied, the headline finding of the entire project emerges. Nominal TRY figures suggest support grew 79% between 2020 and 2023. Inflation-adjusted figures show the opposite: real support fell 43%. In US Dollar terms, the decline was 47%.

Measurement	2020 Total	2023 Total	Change
Nominal TRY	₺27.9 billion	₺50.0 billion	+79%
Real 2023 TRY	₺88.3 billion	₺50.0 billion	-43%
USD equivalent	\$3.97 billion	\$2.11 billion	-47%

This is the most important analytical finding to emerge from the cleaning process, and it would have been invisible without the currency adjustment.

Problem 4: A Suspicious Repeating Value

One specific figure, 703,716,490, appeared in 72 out of 224 rows (32.1% of the dataset). It showed up across multiple different programs, multiple years, and multiple fuel types. In some cases the value was genuinely reported (for programs that carry a fixed nominal value each year, such as certain tax exemption estimates). In other cases, the same figure appeared in 2023 rows where the program's actual value in prior years was a few million lira , suggesting the figure may have been used as a placeholder when the real 2023 data was not yet available at the time the dataset was compiled.

How this was handled

Each instance of 703,716,490 was investigated in context. Where the program consistently carried this value across all four years with zero year-on-year change, it was treated as a genuine fixed estimate. Where the value appeared anomalous relative to the program's history, it was flagged in the data profile sheet for further investigation. Analysts should not use these flagged rows in trend analysis without first verifying the figures against the original source.

Problem 5: Extreme Year-on-Year Changes

Several programs showed year-on-year changes so large that they required individual investigation before being included in any aggregate analysis:

Program	Year	Change	Likely explanation
TUR_TE_17 (VAT exemption for investment machinery)	2022 → 2023	+3,655%	Genuine policy expansion or possible data anomaly , flagged for verification
TUR_TE_16 (Withholding tax support for investments)	2021 → 2022	+360,780%	Driven by placeholder value replacing a small actual figure , flagged
TUR_TE_04 (Oil & gas exploration exemption)	2021 → 2022	+12,718%	Ukraine war shock inflated the value of exploration exemptions , plausible but flagged
TUR_TE_20 (R&D personnel tax support)	2020 → 2021	+4,698%	Likely reflects a real policy change but requires source verification

These outliers do not mean the data is wrong , they mean they need extra scrutiny. An analyst who spots and flags these numbers looks credible. An analyst who quietly includes them in a summary without comment does not.

Problem 6: The Fuel vs Oil Distinction

The dataset contains five fossil fuel type categories: Oil, Fuel, Coal, Natural Gas, and General Fossil Fuel Support. A natural instinct is to merge the Fuel and Oil categories, since both involve petroleum products. This would be a mistake.

- Fuel refers to refined petroleum products at the consumer end of the supply chain: aviation fuel, diesel, naphtha, LPG used in aerosol production, and similar items. The exemptions here reduce the cost of consuming these products.
- Oil refers to upstream activity: exploration rights, pipeline transport of crude oil and gas, and related infrastructure. The exemptions here reduce the cost of finding and moving oil before it is refined.

Merging these categories would mix two entirely different policy instruments aimed at different parts of the supply chain. They were kept strictly separate throughout the analysis.

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Opportunities

The data cleaning process revealed several analytical opportunities that were not visible from the raw data:

Opportunity 1: Tell the Real Story on Spending Trends

The nominal figures suggest fossil fuel support was growing. The inflation-adjusted figures reveal it was shrinking dramatically in real terms. This counterintuitive finding, that Turkey was actually pulling back on real support even as nominal figures rose, is the kind of insight that distinguishes a strong analytical report from a simple data summary. The cleaning process made this finding visible.

Opportunity 2: Pareto Analysis of the Top Programs

With 26 programs consolidated into 104 clean records, it is now straightforward to identify which programs account for the majority of spending. The top three programs alone, the naphtha/petroleum coke exemption (TUR_TE_12), the oil and gas exploration exemption (TUR_TE_04), and the aviation fuel exemption (TUR_TE_07), account for approximately 47% of all support across the four years. Classic Pareto concentration: a few programs doing most of the work.

Opportunity 3: Fuel Type versus Support Type Matrix

Cross-tabulating fuel type against support type (Tax Relief vs Other Support Type) reveals a clear policy pattern. Coal receives both direct cash support and tax breaks. Natural Gas receives almost exclusively direct support, largely through capacity mechanism payments to BOTAŞ. Refined fuel (the Fuel category) is dominated by tax exemptions. This matrix tells a different policy story for each fuel and is only visible after the data is properly aggregated.

Opportunity 4: Outlier Flagging Adds Credibility

By systematically identifying and flagging the suspicious values and extreme year-on-year changes described in the previous section, the cleaned dataset becomes auditable. A reader can see exactly which figures were accepted as-is, which were recalculated, and which require further verification. This transparency strengthens the credibility of any conclusions drawn from the data.

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Recommendations

Based on the issues and opportunities identified during data cleaning, the following recommendations apply to both the analysis phase and any future data collection:

For the Current Analysis

- Always present three versions of financial figures: nominal TRY, real 2023 TRY, and USD equivalent. Never mix these in the same chart or table.
- Lead the written report with the real-terms finding: support fell 43% in purchasing-power terms between 2020 and 2023 despite a 79% nominal increase. This is the headline.
- Use the aggregated 104-row master table (not the raw 224-row file) as the basis for all charts and calculations.
- Do not use the pre-computed YoY_Change, Monetary_Impact, Average_Annual_Support, or Trend columns from the original file. Use the independently recalculated versions.
- Clearly label flagged rows in any presentation or dashboard so that readers know which figures are confirmed and which are pending verification.
- Keep the Fuel and Oil categories strictly separate throughout all analysis.

For Future Data Collection

- Document the methodology behind any pre-computed columns at the time of calculation, including the thresholds used for categorical labels like Monetary Impact.
- Replace placeholder values with clearly coded missing data indicators (such as a blank cell or a standard code like 'N/A') rather than repeating a round number that looks like real data.
- Include a source reference column indicating where each figure came from (budget document, OECD tracker, ministry report) to enable future verification.
- Record the USD/TRY exchange rate and relevant CPI index at the time of data collection so that currency conversions can be reproduced exactly in future updates.

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Data Cleaning Process and Analysis

Step 1: Initial Data Profiling

Before making any changes, the full dataset was inspected to understand its structure. The following was confirmed:

- 224 rows across 17 columns
- 4 years (2020–2023) with exactly 56 rows per year , perfectly balanced
- 26 unique instrument codes representing distinct support programs
- 5 fossil fuel type categories and 2 support type categories
- 26 missing values, all in the YoY_Change column, all corresponding to base-year records where no prior year exists for comparison , this is expected and correct
- Zero nulls in any financial column , all rows had a numeric observation value
- One suspicious repeating value (703,716,490) appearing in 32.1% of rows

Step 2: Sub-Record Investigation and Aggregation

Each instrument code was examined to determine whether multiple rows in the same year represented genuine sub-records or true errors. The review process checked:

- Whether the sum of sub-records produced a round or logical total
- Whether the Instrument Description matched across sub-records
- Whether year-on-year patterns were consistent after aggregation

All 188 multi-row groups were confirmed as legitimate sub-records and were summed to produce a single figure per program per year. The resulting master table has 104 rows.

Step 3: Currency Conversion Setup

An Assumptions tab was created as the single reference point for all currency inputs. The following values were documented with sources:

Year	Annual CPI	Deflator (to 2023 TRY)	USD/TRY Rate
2020	~12.3%	2.70	7.01
2021	19.6%	2.25	8.85
2022	72.3%	1.47	16.55
2023	53.9%	1.00	23.70

Three money columns were created in the master table: Nominal TRY (as reported), Real 2023 TRY (nominal multiplied by the deflator), and USD (nominal divided by the USD/TRY exchange rate). All three columns appear side by side so comparisons across years can be made correctly.

Step 4: Column Recalculation

YoY_Change was recalculated from scratch using the aggregated nominal values, with base-year rows left blank. The formula used was straightforward: $((\text{current year value} - \text{prior year value}) \div \text{prior year value}) \times 100$.

Monetary Impact was re-assigned using documented thresholds: values above ₺1 billion were labelled High Impact; values between ₺100 million and ₺1 billion were labelled Medium Impact;

values below ₺100 million were labelled Low Impact. These thresholds were applied consistently to the aggregated nominal figures.

Trend was recalculated on both the nominal and real series. A program was labelled Increasing if its 2023 value (in the relevant currency) exceeded its 2020 value by more than 5%. It was labelled Decreasing if it fell by more than 5%. Otherwise it was labelled Stable.

Step 5: Outlier Identification and Flagging

After recalculation, records with year-on-year changes exceeding 1,000% were automatically highlighted in the master table. Each was reviewed individually:

- TUR_TE_17 (VAT exemption, investment machinery): A genuine 3,655% increase from 2022 to 2023. The raw sub-record values confirm this jump is present in the source data. It may reflect a real policy expansion following the post-election shift to more orthodox economic policy in mid-2023. Flagged for source verification.
- TUR_TE_16 (Withholding tax, investment certificates): An apparent 360,780% jump driven by the substitution of the placeholder value 703,716,490 for a genuine prior-year figure. This record should not be used in trend analysis until the actual 2022 and 2023 figures are confirmed.
- TUR_DT_07 (Personal income tax exemption for underground workers): Multiple sub-records in 2023 carry the placeholder value, inflating the calculated year-on-year change to implausibly high levels. Flagged throughout.

Step 6: Fuel vs Oil Confirmation

Each occurrence of the Fuel and Oil categories was reviewed against the program description to confirm the distinction. All fuel-category programs described exemptions for refined products consumed at the end of the supply chain. All oil-category programs described exemptions or support for upstream exploration and transport. No cross-category misclassification was found.

Step 7: Data Profile Documentation

A Data Profile tab was created in the workbook documenting every decision made during the cleaning process: which rows were aggregated, which columns were recalculated, which values were flagged, and what thresholds were applied. This serves as the audit trail for anyone who reviews the analysis.

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Conclusion

The raw dataset, while clean in a technical sense (no corrupted files, no missing financial values, no encoding errors), contained three structural challenges that would have produced incorrect analytical conclusions if left unaddressed: the sub-record aggregation issue, the unreliable pre-computed columns, and the nominal currency trap.

After completing the cleaning process, the dataset is now reliable for analysis. The 104-row master table provides one verified figure per program per year in three currency versions, with all derived columns independently recalculated and all flagged values clearly identified.

The most important outcome of this process is not the cleaned table itself , it is the analytical insight that would have been invisible without it. In nominal Turkish Lira, total fossil fuel support appears to have grown by 79% between 2020 and 2023. Once adjusted for inflation, that same support declined by 43% in real terms. Turkey’s government was providing dramatically less real purchasing power in fossil fuel support in 2023 than it was in 2020, despite spending more lira. This is the story the data actually tells, and it is only visible because the currency was handled correctly.

Key takeaway for stakeholders

Do not interpret rising nominal figures as evidence of growing government commitment to fossil fuel support. The lira’s collapse means that every lira allocated in 2023 was worth significantly less than a lira allocated in 2020. The real story in this dataset is one of declining real-terms support , even as the nominal numbers point upward. All charts and summaries in this project should make this distinction visible.

The cleaned dataset, the assumptions documentation, and the data profile are all available in the accompanying workbook. Any further queries about specific cleaning decisions should be referred to the Data Profile tab in that workbook.